COURSE CODE/TITLE: CPE 010-CPE21S1 - Data Structures and Algorithms	**SECTION: CPE21S1**
DATE PERFORMED: July 16 2026	**DATE SUBMITTED: July 21 2026**
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ACTIVITY NO.2.1 **Arrays, Pointers and Dynamic Memory Allocation**

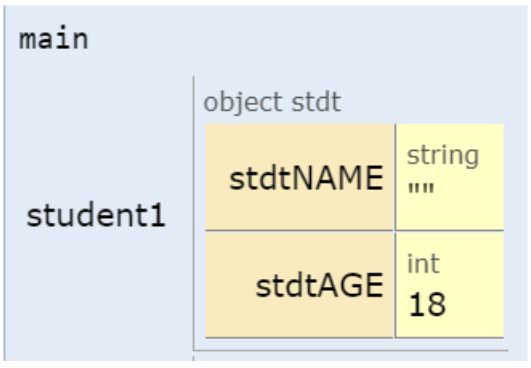
1. PROCEDURE OUTPUT

ILO A: Implement static and dynamic memory allocation & ILO B: Create dynamically allocated objects using pointers and array

Screenshot

Print output (drag lower right corner to resize)

```
The CONSTRUCTOR was CALLED
Copy Constructor was called
The CONSTRUCTOR was CALLED
The DESTRUCTOR was CALLED
The DESTRUCTOR was CALLED
The DESTRUCTOR was CALLED
```



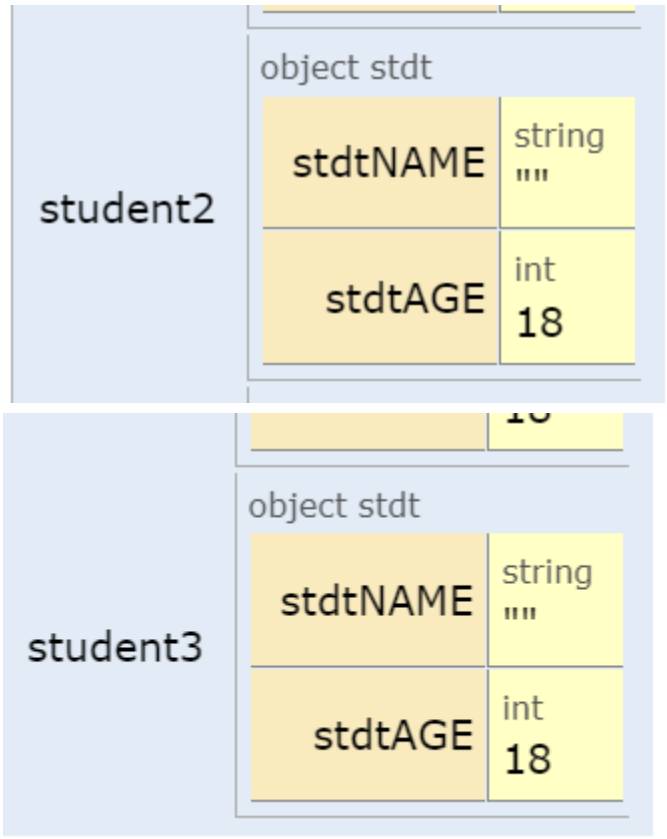
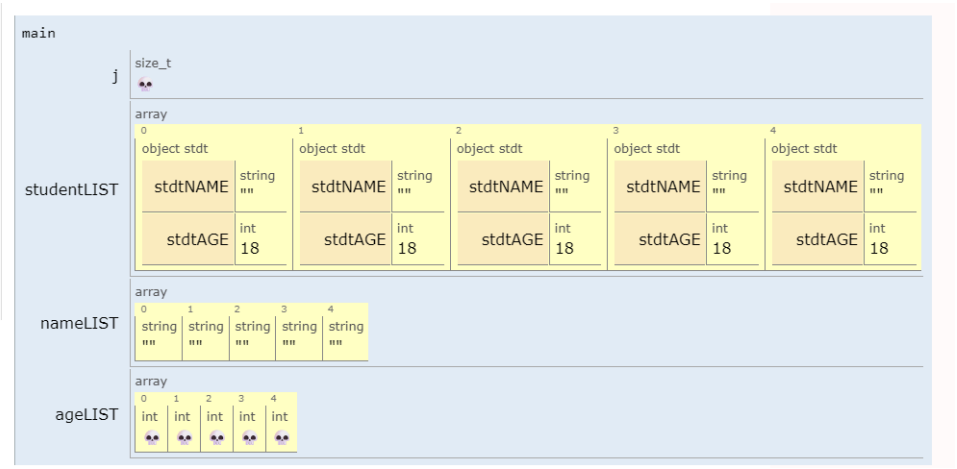
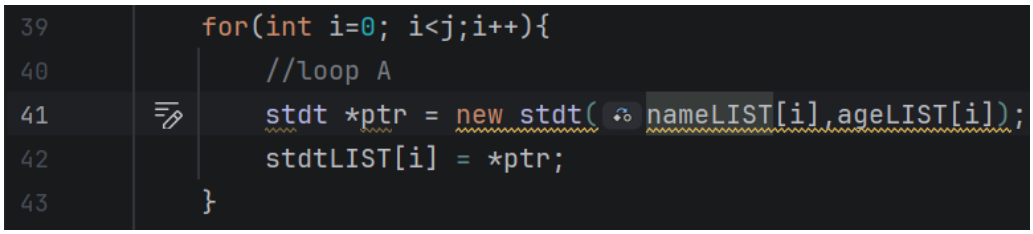
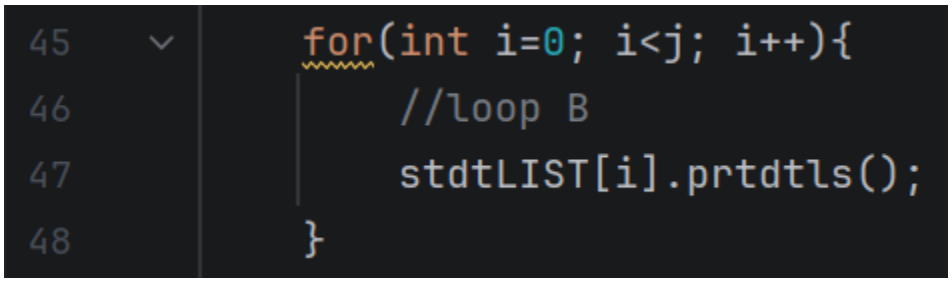
	
Observation	FIRST PART: All student(**stdt**) objects(the **student1, student2,** and **student3)** are declared as local variables inside of the **main()** without using of the **new,** which means, they are only at the stack and not using or going to the heap. The De allocation happens automatically after each of the **student object(1 to 3)** in reverse, this shows that it is a static memory allocation being automatically done by the compiler.

Table 1.1: Initial Driver Program

Screenshot	

	Print output (drag lower right corner to resize) The CONSTRUCTOR was CALLED The CONSTRUCTOR was CALLED The CONSTRUCTOR was CALLED The CONSTRUCTOR was CALLED The CONSTRUCTOR was CALLED The DESTRUCTOR was CALLED The DESTRUCTOR was CALLED The DESTRUCTOR was CALLED The DESTRUCTOR was CALLED The DESTRUCTOR was CALLED
Observation	SECOND PART: The modified code uses arrays or shows the use of stack/ static memory allocation with the use of arrays by declaring the **studentLIST** an array with the size of 5 using the declaration **const size_t j=5**; to define the array size. The constructor was called 5 times, for every name included on the array(along with the age), as well as the destructor for finishing up the freeing up the resources.

Table 1.2. Modified Driver Program with Student Lists

Loop A	
Observation	The loop A was able to use the **new** keyword to store it or make it into the heap using dynamic allocation, for every iteration(5 times). This called the constructor 10 times since there are now data or values to be placed or passed in(the names and age). The **ptr** created then used to copy the data into the **stdLIST[i]** so that the values/data from the **nameLIST and ageLIST** moved into the array.
Loop B	
Observation	Loop B, like loop A, called out the **prtftls()** for every student object in order to display the name and age. Since loop A already added the data inputted into the array using the copy assignment,

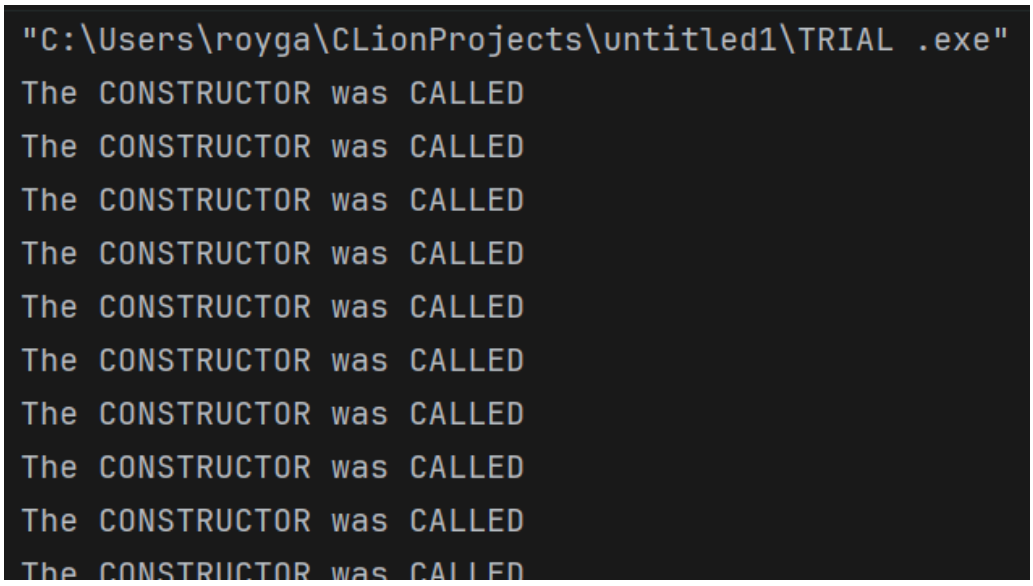
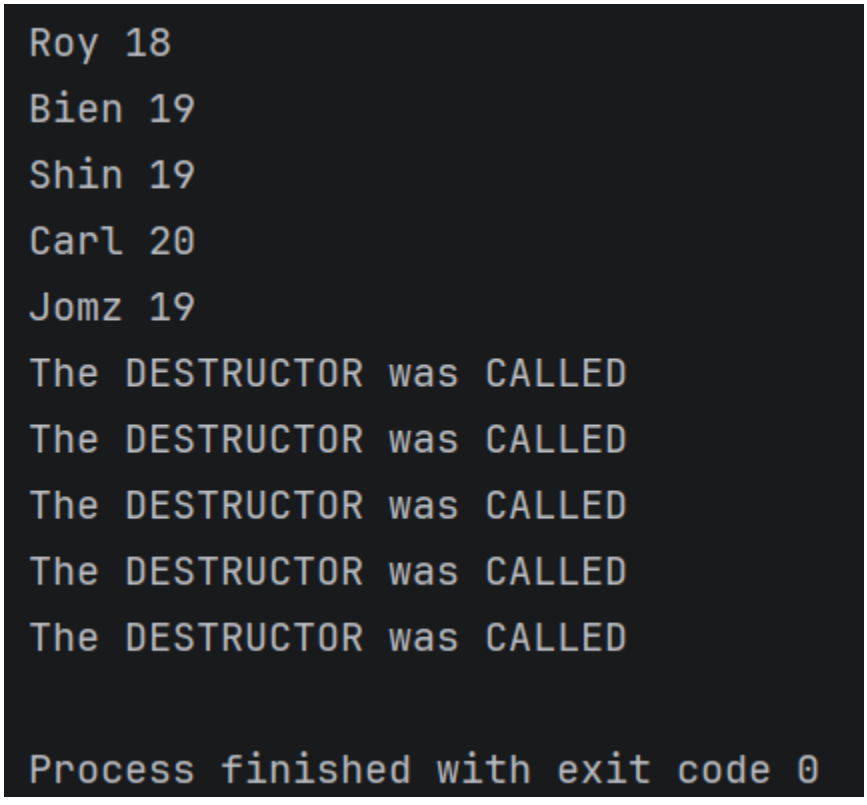
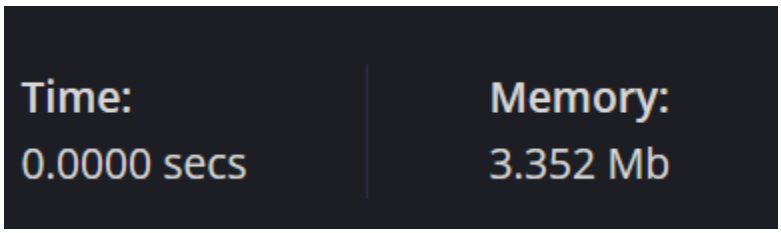
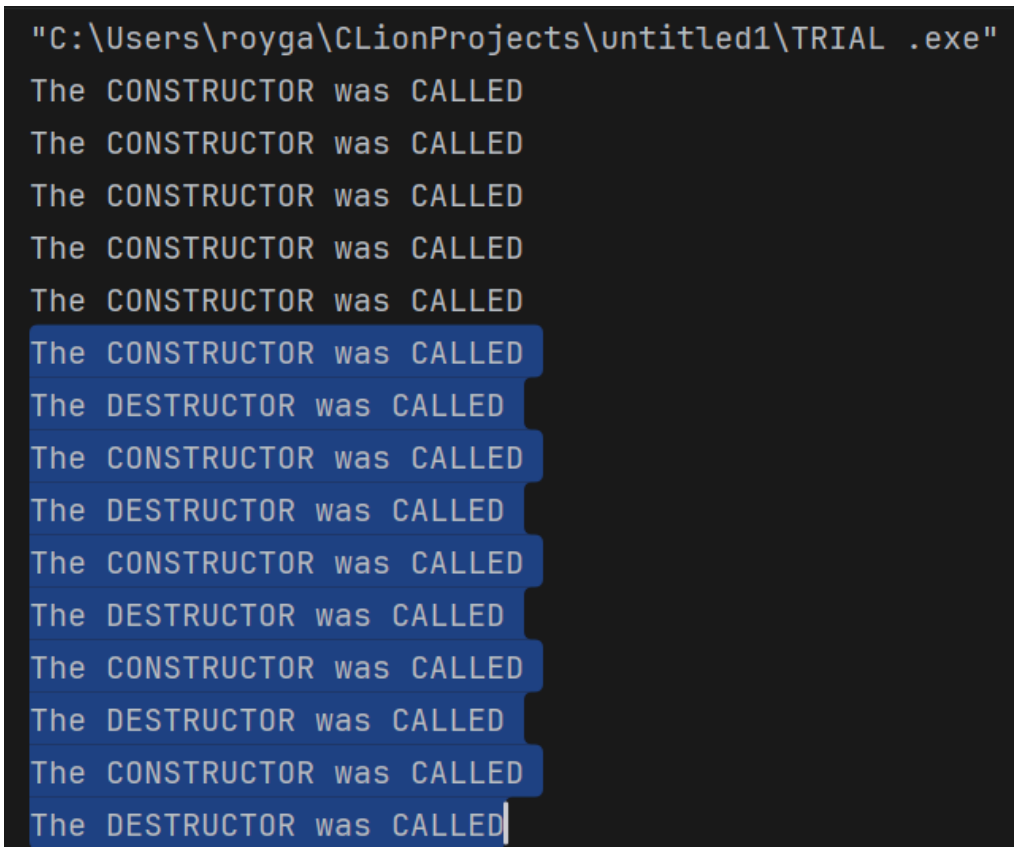
	the values that are printed out are not the John Doe's and age 18.
Output	 
Observation	

Table 1.3. Final Driver Program

Modifications	  

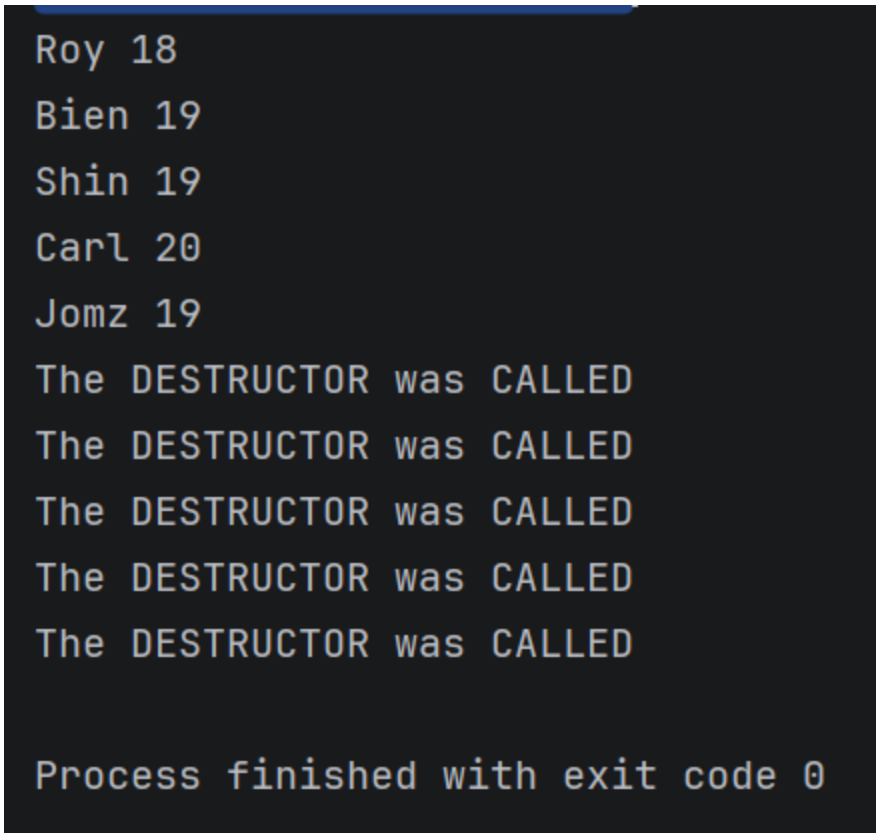
	
Observations	The modifications that I've done is just following the AUDNu, specifically just deleting the pointers to add up more resources for the user. Basically stopping a "memory Leak".

Table 1.4. Modifications/Corrections Necessary

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Questions/Task:

ILO C: Solve programming problems using dynamic memory allocation, arrays and pointers

Jenna's Grocery List		
Apple	PHP 10	x7
Banana	PHP 10	x8
Broccoli	PHP 60	x12
Lettuce	PHP 50	x10

Jenna wants to buy the following fruits and vegetables for her daily consumption. However, she needs to distinguish between fruit and vegetable, as well as calculate the sum of prices that she has to pay in total.

Problem 1: Create a class for the fruit and the vegetable classes. Each class must have a constructor, deconstructor, copy constructor and copy assignment operator. They must also have all relevant attributes (such as name, price and quantity) and functions (such as calculate sum) as presented in the problem description above.

Problem 2: Create an array GroceryList in the driver code that will contain all items in Jenna's Grocery List. You must then access each saved instance and display all details about the items.

Problem 3: Create a function TotalSum that will calculate the sum of all objects listed in Jenna's Grocery List.

Problem 4: Delete the Lettuce from Jenna's GroceryList list and de-allocate the memory assigned.

Problem 5: Discuss the

Output:

```
"C:\Users\royga\CLionProjects\untitled1\ILO C  ACT 2.1.exe"
Constructor was Called Apple
Constructor was Called Banana
Constructor was Called Brocoli
Constructor was Called Lettuce
```

```
|--Jenna's Grocery List--|

FRUITS:
[Fruit] Apple | PHP 10 x 7 = PHP 70
[Fruit] Banana | PHP 10 x 8 = PHP 80

VEGETABLES
[Vegetable] Brocoli | PHP 60 x 12 = PHP 720
[Vegetable] Lettuce | PHP 50 x 10 = PHP 500
Destructor was Called Lettuce

VEGETABLES AFTER DELETION
[Vegetable] Brocoli | PHP 60 x 12 = PHP 720
```

```
========================================
     UPDATED RECEIPT
========================================

-- FRUITS --
[Fruit] Apple | PHP 10 x 7 = PHP 70
[Fruit] Banana | PHP 10 x 8 = PHP 80

-- VEGETABLES --
[Vegetable] Brocoli | PHP 60 x 12 = PHP 720
========================================
UPDATED TOTAL: PHP 870
========================================
Destructor was Called Brocoli
Destructor was Called Banana
Destructor was Called Apple

Process finished with exit code 0
```

CONCLUSION:

Provide the following:

1. Summary of lessons learned
2. Analysis of the procedure
3. Analysis of the supplementary activity
4. Concluding statement / Feedback: How well did you think you did in this activity? What are your areas for improvement?

1. Through this activity, I was able to have a proper code(I do not know if this is the standard way) that have the 4 basic “special member function”? I was also able to see the procedure that the code is following through the use of code visualizer and now understand the process, but would need to use it again for other activity.

2. The procedure was easy enough with the us of code visualizer and references from sir and geekforgeeks, it showed on when and where did the special members fire and what is the content of them in the stack and heap.

3. This one is hard, took me 2 days actually(around 12 hours total), I was able to 2 code(1 is like a simple but would not work, and needed to be hardcoded for other part), but only one was acceptable for the requirement of the task. Focussing on the proper and working, I was able to easily make the class and functions like the fruits and vegetable class, and the calculator function(of course some part of the main), The problem that I had was the code for the destructor in the heap, specifically to remove the lettuce vegetable and make it update in the final list, so I did use AI to help me, but it was not literally copy paste. Furthermore, since the visualizer is not working for this long code(did not run due to the length) I do not truly have the visual analysis on how the code worked in the heap, just the analyzation from the code running.

4. The first activity was easy enough, since the code was basically provided already, if we will have to do it again but without, It would take time, but doable, Now for the Supplementary, I truly have to redo? Or review once again the special members on how to do them(not the address parts) since that is where I am lacking, I know the basics and theory, but have a hard time to apply it into practice

4. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Claude Ai**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Formatting and checking the grammar, also used to supplement the code**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Word phrasing and proper formatting, this is basically just to speed up the formatting. Additionally, for the deletion of the heap array(vegetable) since the user's original code was not working(specifically for the lettuce)**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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